

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
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Section / Batch:	AI&DS 2 nd Year	Date:	22/07/2026

Code of Conduct

I R.Rohini

Sree(192524471)_____

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R.Rohini sree_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marks

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

```
1 #include<stdio.h>  
2 int stack[100];  
3 int top=-1;  
4 int pop() {  
5     if(top==0){  
6         printf("under flow");  
7         return -1;  
8     }  
9     return stack[top--];  
10 }
```

10

20

30

Output

30

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int stack[5];
3 int top;
4 void init() {
5     top=-1;
6 }
7 int main() {
8     init();
9     printf("top=%d",top);
10    return 0;
11 }
```

Test Input

stdin input (optional)

Output

top=-1

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marks

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
    while(stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (ch==' '){
2     while (stack[top]!=' '){
3         postfix[j++] = pop();
4     }
5     pop();
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1**
Binary Search
5 marks
- Q2**
Stack
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Stack
5 marks
- Q6**
Linked List
5 marks
- Q7**
Stack
5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[') {
    pop();
} else {
    return 0;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((ch=='[' && stack[top]=='[') || (ch==']' && stack[top]==')) {
2     pop();
3 }
4 else{
5     return 0;
6 }
```

Test Input

stdin input (optional)

Output
QUESTIONS

- Q1**
Binary Search
5 marks
- Q2**
Stack
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Stack
5 marks
- Q6**
Linked List
5 marks
- Q7**
Stack
5 marks

Q6 Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
temp = temp.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp=head
2 while temp!=null:
3     print(temp.data)
4     temp=temp.next
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack
5 marks
- Q6
Linked List
5 marks
- Q7
Stack
5 marks

Q7 Stack

5 marks

Point out the error.

```
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack,top):
2     if top==0:
3         print("under flow")
4         return stack,top
5     else:
6         item=stack[top]
7         top-=1
8         return item,top
```

Test Input
 stdin input (optional)

Output
QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack
5 marks
- Q6
Linked List
5 marks
- Q7
Stack
5 marks

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
current.next = prev
prev = current
current = current.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 prev=NULL
2 while current!=null:
3     next_node=current.next
4     current.next=prev
5     prev=current
6     current=next_node
```

Test Input
 stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marks

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1. newNode.next=head;  
2. head=newNode;
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1. DEQUEUE(Q):  
2.     if front == -1:  
3.         print("underflow")  
4.     else:  
5.         temp=Q[front]  
6.         if front==rear:  
7.             rear=-1  
8.         else:  
9.             front=front+1
```

Test Input

stdin input (optional)

Output

CO2_A12_DEBUGGING & OPTIMIZATION TASK (SET 2)

BACK

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7****Q1** Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while(expression[i]!='\0'){  
2     if(isdigit(expression[i]))  
3         push(expression[i]-'0');  
4     else{  
5         int op2=pop();  
6         int op1=pop();
```

Test Input

stdin input (optional)

Output

CO2_A12_DEBUGGING & OPTIMIZATION TASK (SET 2)

BACK

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List**Q2** Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {  
2     if (front == -1)  
3         return -1;  
4     int data = queue[front];  
5     if (front == rear){  
6         front = rear = -1;  
7     } else {
```

Test Input

0

Output

CO2_A12_DEBUGGING & OPTIMIZATION TASK (SET 2)

BACK

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7****Q3 Binary Search**

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarysearch(int arr[],int l,int r,int x){  
2     if(l>r)  
3         return -1;  
4     int mid=l+(r-l)/2;  
5     if(arr[mid]==x)  
6         return mid;  
7     if(arr[mid]>x)  
8         return binarysearch(arr,l,mid-1,x);  
9     return binarysearch(arr,mid+1,r,x);  
}
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7****Q4 Queue**

5 marks

Find the error in this Deque InsertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data){  
2     if((front==0&&rear==MAX-1)|| (front==rear+1)){  
3         printf("Deque full");  
4         return;  
5     }  
6     if(front==0){  
7         front=rear;  
8     }  
9     else if(front==0){  
10        front=MAX-1;  
11    }else{
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (SET 2)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {  
2     if (rear==MAX-1) {  
3         return;  
4     }  
5     queue[++rear]=data;  
6 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (SET 2)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deletenode(node**head,node*del) {  
2     if (*head==del)  
3         *head=del->next;  
4     if (del->next!=null)  
5         del->next->prev=del->prev;  
6     if (del->prev!=null)  
7         del->prev->next=del->next;  
8     free (del);  
9 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1**
Stack Notations - Postfix Expression
5 marks
- Q2**
Circular Queue
5 marks
- Q3**
Binary Search
5 marks
- Q4**
Queue
5 marks
- Q5**
Queue
5 marks
- Q6**
Doubly Linked List
5 marks
- Q7**
Stack
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP():  
2 if(top==0)  
3   underflow;  
4 else  
5   return stack[top--];
```

Test Input

stdin input (optional)

Output



Pending manual review

Hot days ahead

Search

ENG

QUESTIONS

- Q1**
Stack Notations - Postfix Expression
5 marks
- Q2**
Circular Queue
5 marks
- Q3**
Binary Search
5 marks
- Q4**
Queue
5 marks
- Q5**
Queue
5 marks
- Q6**
Doubly Linked List
5 marks
- Q7**

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(top==MAX-1){  
2   print("overflow");  
3 }  
4 else{  
5   stack[++top]=item;  
6 }
```

Test Input

stdin input (optional)

Output



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q9** Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;  
queue[rear] = vehicle;  
if(rear == front){  
    printf("Queue Full");  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((rear+1) % MAX == front) {  
2     printf("Queue Full");  
3 }  
4 else{  
5     rear=(rear+1)%MAX;  
6     queue[rear]=vehicle;  
7 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q10** Stack

5 marks

Point out the error?

```
PEEK(stack):  
return stack[top+1]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 PEEK(stack):  
2     return stack[top]
```

Test Input

stdin input (optional)

Output